



Air Quality & Odour Assessment

Post Mining Land Use Pilot Project – Macquarie Coal Complex (the Rezoning Proposal)

Department of Planning, Housing and Infrastructure

Prepared by:

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Basis of Report

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Executive Summary

The Macquarie Coal Complex (MCC) Pilot Project was initiated as part of the NSW Government's wider response to a parliamentary inquiry into beneficial and productive post-mining land use. In July 2025, the NSW Government committed to the initiation and development of two pilot projects to showcase post-mining land use benefits and foster improvements for achieving full relinquishment and redevelopment of the mine sites.

The MCC Precinct ("the Precinct") rezoning was prepared by the Department of Planning, Housing and Infrastructure (DPHI) in partnership with Lake Macquarie City Council, Net Zero Economy Authority and NSW Government agencies. The DPHI and Lake Macquarie City Council have undertaken master planning informed by technical studies, to support coordinated transition from post mining land use to alternative employment generating land uses in the Precinct.

SLR Consulting Australia Pty Ltd (SLR) was engaged by Aurecon Pty Ltd (Aurecon) on behalf of the DPHI to undertake an Air Quality and Odour Assessment to inform planning considerations associated with the proposed state-led rezoning of the Precinct.

The Air Quality (including odour) Impact Assessment (AQIA):

- Provides a strategic air quality and odour evidence base to inform land use compatibility and buffer principles in the MCC Pilot Project 2026 ("the Master Plan").
- Responds to the Enquiry by Design-identified context that air pollutant emissions will arise from construction, industrial operations, legacy/active coal-related activities and transport corridors, and that local meteorology is complex.
- Defines performance criteria and triggers for detailed assessment in later phases, following rezoning.

The general purpose of an AQIA is to determine and test the potential for air pollution impacts on:

- Human health e.g. due to high, measured or predicted concentrations of air pollutants drawn into the human lung.
- Flora and fauna including the effects of air pollutant deposition on land, rivers, coasts, and the sea.
- Amenity, caused by:
 - o odour emissions.
 - o dust deposition on surfaces such as clothes, dwellings, and vehicles.
 - o visible plumes, and more widely, visibility.

The primary constraints imposed by any AQIA are air emissions limits (for the point of emission), and ambient air quality standards for the surrounding environment.

This AQIA is a strategic, desktop-based evaluation of current and potential future air quality and odour impacts within the Precinct to inform land-use planning, identify development opportunities and constraints, and recommend best-practice management approaches to support future planning controls.

The assessment focusses on regional context, existing and potential future air pollution sources, pathways, receptors, risks, and planning-scale recommendations.

The Precinct will need to minimise environmental impacts due to air pollutant emissions. Air quality impacts within the Precinct can be effectively managed by, where needed, monitoring



and detailed (modelling) investigations on a case-by-case basis, and environmental (air quality) management plans for larger projects – in addition:

- The rehabilitation program to stabilise and cover a tailings storage facility is very important. More generally, revegetation and stabilisation of exposed surfaces in the Precinct will reduce the potential for dust emissions. Vegetation acts as a wind break, dust sink, and assists with erosion control.
- The environmentally constrained areas will provide vegetated corridors between new land uses and air pollutant emission sources such as industries and roads, reducing dust transport and improving air pollutant dispersion.
- Integration of low emissions industries will represent a shift from high emissions to lower air emissions land use.
- Proper planning and design for the Precinct will help to prepare for future climate impacts from global warming such as, potentially, longer bushfire seasons, and more frequent and more intense wind, dust, and precipitation storms.
- Post-mining land uses selected to deliver lower air pollutant emissions for air quality management purposes, are likely to lower greenhouse gas emissions also (a co-benefit).

This desktop AQIA was limited to a review of confidential Project material and readily available, publicly available information. Exclusions from the scope of works included:

- Meteorological, and air pollutant/odour monitoring.
- Air pollutant deposition monitoring.
- Meteorological and air pollutant/odour dispersion modelling.

The AQIA is intended for strategic planning purposes. It does not replace project-specific, air quality/odour impact assessment that may be required in a later stage of development and/or approval.

All future Development Approvals, re-zoning, and change of land use are to be accompanied by proportional and site-specific air quality and/or odour impact assessments as required. Additionally:

- An airborne particulate / deposited dust monitoring program in the Precinct may be useful to demonstrate improved mitigation of tailings dust emissions.
- More generally, revegetation and stabilisation of exposed surfaces in the Precinct will reduce the potential for dust emissions – vegetation acts as a wind break, a dust sink, and assists with erosion control. Use of native vegetation should increase resilience of exposed surfaces to wind erosion.

If possible, legacy and / or existing air monitoring infrastructure could be used to establish a new monitoring network, providing a new baseline for future development in the Precinct.



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Abbreviations and Definitions

Abbreviation	Expansion / Definition
AQIA	Air Quality Impact Assessment – for this report, odour impact assessment falls within the scope of AQIA.
Aurecon	Aurecon Pty Ltd.
CASANZ	Clean Air Society of Australia and New Zealand
CAQP	Certified Air Quality Professional
CO	Carbon monoxide (not the greenhouse gas, carbon dioxide).
DCCEEW	Department of Climate Change, Energy, the Environment and Water (Australian Government).
DPHI	Department of Planning, Housing and Infrastructure
Engine	Fossil fuel-powered combustion engine e.g. gas turbine, reciprocating internal combustion engine.
EPA	Environment Protection Authority (NSW).
Fossil fuel	Coal, natural gas (methane), diesel, petrol, liquefied natural gas, liquefied petroleum gas.
µg; µg/m ³	microgram; microgram per cubic metre
µm	micrometre (1/1000 th of a millimetre).
MCC	Macquarie Coal Complex
Master Plan	Macquarie Coal Complex Pilot Project Master Plan 2026
NO	Nitric oxide (see also NO ₂ and NO _x).
NO ₂	Nitrogen dioxide – a product of the combustion of fossil fuels, being a component of NO _x . Typically, in engine exhaust, approximately 10% of NO _x emissions may be NO ₂ (approximately 90% being NO). In sunlight, NO ₂ can photolyse to form NO and an oxygen atom (O). The oxygen atom may form O ₃ by reacting with molecular oxygen (O ₂).
NO _x	Oxides of nitrogen, primarily NO and NO ₂ .
NPI	National Pollutant Inventory
NSW	New South Wales
O ₃	Ozone (see also NO ₂).
Odour	Odorous, gaseous molecules e.g. hydrogen sulfide, or a complex mixture of odorous molecules e.g. mercaptans.
PM _{2.5}	Particulate Matter 2.5 – small particles with aerodynamic diameters less than 2.5 µm; concentration usually given in units of “µg/m ³ ”.
PM ₁₀	Particulate Matter 10 – small particles with aerodynamic diameters less than 10 µm; concentration usually given in units of “µg/m ³ ”.
PMLU	Post Mining Land Use
ppb	Parts per billion (volumetric concentration; for gases).
ppm	Parts per million (volumetric concentration; for gases).
Precinct	Macquarie Coal Complex Precinct



Abbreviation	Expansion / Definition
SEPP	<i>State Environmental Planning Policy (Precincts – Regional) 2021 (Precincts – Regional SEPP)</i>
Site	Macquarie Coal Complex (the Precinct) and Teralba Southgate.
SLR	SLR Consulting Australia Pty Ltd.
SO ₂	Sulfur dioxide; a significant air emission from coal combustion.
TSF	Tailings Storage Facility
VOC	Volatile Organic Compounds – for NPI reporting, chemical compounds based on carbon chains or rings, with a vapour pressure greater than 0.01 kiloPascal at a temperature of 293.15 Kelvin (20°C), that participate in atmospheric photochemical reactions – exclusions include CO and methane (DCCEEW, 2026).
WTW	Wastewater Treatment Works



1.0 Introduction

1.1 Background

The Macquarie Coal Complex (MCC) Pilot Project was initiated as part of the NSW Government's wider response to a parliamentary inquiry into beneficial and productive post-mining land use. In July 2025, the NSW Government committed to the initiation and development of two pilot projects to showcase post-mining land use benefits and foster improvements for achieving full relinquishment and redevelopment of the mine sites (DPHI, 2026).

The MCC ("the Precinct") rezoning was prepared by the Department of Planning, Housing and Infrastructure (DPHI) in partnership with Lake Macquarie City Council, Net Zero Economy Authority and NSW Government agencies.

The DPHI and Lake Macquarie City Council have undertaken master planning informed by technical studies, to support coordinated transition from post mining land use to alternative employment generating land uses in the Precinct (DPHI, 2026).

1.2 Assessment Overview

SLR Consulting Australia Pty Ltd (SLR) has been engaged by Aurecon Pty Ltd (Aurecon) on behalf of the DPHI to undertake an Air Quality and Odour Assessment to inform planning considerations associated with the proposed state-led rezoning of the Precinct.

1.3 Purpose

The purpose of this Air Quality (including odour) Impact Assessment (AQIA) is to:

- Provide a strategic air quality and odour evidence base to inform land use compatibility and buffer principles in the MCC Pilot Project 2026 ("the Master Plan").
- Respond to the Enquiry by Design (EbD)-identified context that air pollutant emissions will arise from construction, industrial operations, legacy / active coal-related activities and transport corridors, and that local meteorology is complex.
- Define performance criteria and triggers for detailed assessment in later phases, following rezoning.

1.4 AQIA Definition

The general purpose of an AQIA is to determine and test the potential for air pollution impacts on:

- Human health e.g. due to high, measured or predicted concentrations of air pollutants drawn into the human lung.
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- Amenity, caused by:
 - o odour emissions.
 - o dust deposition on surfaces such as clothes, dwellings, and vehicles.
 - o visible plumes, and more widely, visibility.



The primary constraints imposed by any AQIA are air emissions limits (for the point of emission), and ambient air quality standards for the surrounding environment; these are detailed in the next section.

1.5 Scope of AQIA

This AQIA is a strategic, desktop-based evaluation of current and potential future air quality and odour impacts within the Precinct to inform land-use planning, identify development opportunities and constraints, and recommend best-practice management approaches to support future planning controls.

The assessment is not a detailed air quality and odour impact modelling study, instead focussing on regional context, existing and potential future air pollution sources, pathways, receptors, risks, and planning-scale recommendations.

The objectives are to:

- Identify potential sources of odour and/or air quality impacts generated from the investigation area.
- Develop an understanding of the nature of any odour and significant air pollution sources.
- Desktop review and screening assessment.

Exclusions including monitoring and modelling.

The methodology is detailed in **Section 4.0**.

2.0 Precinct Structure Plan

2.1 Overview

The Site subject to rezoning includes the Precinct, and Teralba Southgate. The Precinct (only) is being rezoned under the *State Environmental Planning Policy (Precincts – Regional) 2021 (Precincts – Regional SEPP)*, and subject to the Master Plan.

The rezoning proposal applies to 1,160 hectares of land and includes land in the Precinct, and Teralba Southgate sub-precinct, across the following areas:

- West Wallsend Underground (ceased mining in 2016).
- Westside Open Cut (ceased mining in 2012).
- Macquarie Coal Preparation Plant (ceased production in 2016).
- Teralba Southgate (ceased mining in 2001).

The Site is illustrated in the Structure Plan provided in **Appendix A**.

The Site is on the land of the Awabakal people, and currently owned by Glencore. All mining operations on the Site have ceased, with the site advancing through rehabilitation works in accordance with the Macquarie Coal Rehabilitation Management Plan (2022). The Site is located in the Lake Macquarie Local Government Area, approximately 20 kilometres south-west of Newcastle and approximately 110 kilometres north of the Sydney CBD.

From the perspective of air quality assessment, the Precinct and Teralba Southgate are characterised by:

- Legacy and active coal mining and coal handling infrastructure;



- Proximity to major regional industrial sources, including coal-fired power generation at Eraring and Vales Point;
- Existing transport corridors; and
- Complex local meteorological conditions, described in **Section 6.2**.

2.2 Structure Plan

The structure plan is the principal spatial framework for the Master Plan, identifying the arrangement of developable areas, environmentally constrained areas, recreational areas, movement networks and key constraints across the Precinct. It is understood the boundaries between land use categories shown on the structure plan are indicative and may be refined through detailed technical investigation and assessment at the development application stage DPHI (2026).

The structure plan, with some labelling added for this AQIA, is provided as an attachment in **Appendix A**). Categories of areas within the structure plan are (DPHI, 2026):

- **Developable areas** – land assessed as most likely to be suitable for development, subject to further technical investigation and site-specific assessment at the development application stage.
 - o Developable areas are concentrated in locations with relative geotechnical stability, accessibility to existing or planned transport networks and proximity to utility infrastructure or feasible servicing solutions.
- **Environmentally constrained areas** – land identified as having high biodiversity values or where natural features (watercourses, wetlands, steep slope, flooding) constrain development for employment uses.
 - o Detailed environmental investigations are required prior to development or disturbance of vegetation in these areas.
 - o Environmentally constrained areas form an interconnected network across the Precinct that underpins ecological connectivity, water cycle management and landscape identity.
- **Recreational areas** – land suitable for active and passive recreation facilities, balanced with protecting biodiversity and cultural values.
 - o Primarily these are associated with the West Lake sub-precinct and the conservation corridor network, where the landscape and environmental values support nature-based tourism, walking and cycling trails and community recreational facilities.
- **Developable land subject to further investigation** – land constrained in the short to medium term by geotechnical, dam safety, flood or biodiversity factors requiring resolution before development can proceed.
 - o Further investigation is required prior to development.
 - o Release of this land for development is contingent upon completion of technical investigations and the achievement of applicable staging triggers.
- **Consideration areas** – located on interfaces between developable land and sensitive receptors, environmental features or adjacent communities; these are illustrated in **Appendix B** – the same plan used by the SLR (2026) noise and vibration assessment).



- o Development within or adjacent to a consideration area must address the relevant interface typology and demonstrate compliance with performance criteria.
- **Movement network** – existing and proposed road, freight and active transport corridors that structure internal and external connectivity.
- **Key constraints** – significant infrastructure, environmental or hazard features; e.g. TSF, pipeline, coastal wetlands, and transmission lines, that influence land use allocation and development design.

Surrounding Land Uses illustrated are (see **Appendix A**):

- Residential – rural, suburban and future.
- Heavy industry / mining.
- Commercial / light industry.
- Education.
- Recreation.

3.0 Pollutants of Concern

This AQIA identified the primary pollutants of concern for the Precinct as:

- sulfur dioxide (SO₂);
- nitrogen dioxide (NO₂) and ozone (O₃); linked by atmospheric chemistry
- small dust particles assessed as Particulate Matter 2.5 (PM_{2.5}) and Particulate Matter 10 (PM₁₀); (see Abbreviations for definitions);
- deposited dust (the airborne dust mass settled on a land or water surface); and
- individual and complex mixtures of toxic air pollutants and similarly for odorous air pollutants.

4.0 Assessment Methodology

The qualitative air quality and odour assessment methodology is detailed in the following list of tasks:

- Desktop review of general/background information relevant for air quality and odour impacts in the study area e.g.
 - o Air quality/odour studies, mine rehabilitation plans, closure plans, operational environmental plans, existing and proposed land use zoning, precinct plans, strategic planning reports.
 - o Local topographic and meteorological influences on air pollutant dispersion.
- Identification of key air pollutant emission sources relevant for the study area including:
 - o Residual mining operations and mine rehabilitation activities, stockpiles, tailings, coal handling, and waste rock areas.
 - o Legacy infrastructure with potential for dust or odour generation.
 - o Existing/future mobile equipment as diesel combustion sources, and vehicle movements.



- o Potential future development scenarios where indicative land uses are known.
- o Surrounding and regional sources such as active coal mining and infrastructure, quarries, manufacturing, power infrastructure, and waste facilities; and,
- o Major roads and transport corridors, agricultural operations, wood smoke and other domestic sources, and controlled burns and bushfires.
- Characterisation of emissions and impact potential; for each identified source:
 - o Identify and list the critical air pollutants e.g. particulate matter, odour, gaseous criteria pollutant species, gaseous toxic pollutant species.
 - o Characterise emissions by magnitude and temporally where practicable.
 - o Note any intensity changes expected during mine closure, rehabilitation phases, or future redevelopment.
- Review available meteorological controls affecting dispersion, such as temperature inversions, lake breezes and sea breezes, terrain flows.
- Identification of sensitive receptors and planning interfaces.
 - o Map existing sensitive receptors including residences, schools, aged care, health facilities, and recreation areas.
 - o Identify and map future land use opportunities and interface constraints.
 - o Identify and map potential impacts on non-human receptors such as national parks, water bodies, ecological communities.
- Strategic impact evaluation
 - o Using a qualitative / semi-quantitative framework, evaluate potential air quality impacts.
 - o Evaluate potential odour impacts, annoyance risk, and compatibility with future land uses.
 - o Identify development constraints, including required separation distances or buffers.
 - o Identify opportunities for compatible development zones e.g., industry locations shielded from communities, areas with low odour/dust exposure risk.
 - o Highlight high risk exposure pathways e.g., valley trapping, prevailing wind alignment.
 - o Provide relative risk categories and spatial risk mapping (see next point).
- Mapping and visualisation with spatial outputs showing:
 - o Source locations, wind roses, qualitative air pollutant dispersion, significant terrain, and sensitive receptors.
 - o Air quality and odour risk mapping.
 - o Recommended buffer zones or separation distances.
 - o Conceptual land use compatibility overlays.
- Best practice management and strategic planning recommendations.



- o Provide guidance consistent with NSW Environmental Protection Agency (EPA) and global best practice for strategic planning controls, mitigation strategies for dust, odour, and combustion emissions relevant to future development.
- o Recommendations for vegetation buffers and landscape elements, setbacks from industrial areas, staging of rehabilitation or development to minimise exposure, and controls suitable for e.g. precinct plans.
- o Identify data gaps and recommend future monitoring, modelling, or detailed assessments required for subsequent planning stages.
- Reporting.

5.0 NSW Legislation and Guidelines

5.1 Overview

Any specific air quality constraints for the Precinct will be determined by measures set out for the protection of the environment due to air pollution by NSW legislation, with specific methods set out in various guidelines. These are summarised in this section.

5.2 NSW Legislation

The *Protection of the Environment Operations Act 1997* No 156, (“the POEO Act”), regulates pollution in NSW, and is administered by the NSW Environment Protection Authority (EPA). An objective of the POEO Act is to reduce air discharges such that the risks to human health and the environment are “minimal”, or harmless.

5.3 NSW Air Emissions Limits

Air pollutant emissions must not exceed concentration limits prescribed by the *POEO (Clean Air) Regulation 2022* for industrial point (stack) sources. The Regulation’s limits are based on levels achievable through the application of “reasonably available technology and good environmental practices”. In some circumstances, other legal instruments may impose tighter emissions limits.

5.4 NSW Assessment Methods

The NSW Environmental Protection Authority (EPA) *Approved Methods for the Modelling and Assessment of Air Pollutants in New South Wales* (EPA, 2022), lists the statutory methods for modelling and assessing emissions of air pollutants from stationary sources in NSW. In this assessment, consideration was given to draft changes set out in the 2026 draft update (EPA, 2026); public submissions have been completed.

EPA (2022) sets out detailed procedures for:

- Preparation of air emissions and meteorological data;
- Accounting for background (existing) air pollutant concentrations;
- Air pollutant dispersion modelling;
- Modelling of chemical transformation; and
- Interpretation of model results.



5.5 NSW Air Quality Impact Assessment Criteria

Critically, EPA (2022) provides lists of air quality impact assessment criteria for air pollutants and air quality indicators, including:

- sulfur dioxide (SO₂);
- nitrogen dioxide (NO₂) and ozone (O₃);
- small dust particles assessed as Particulate Matter 2.5 (PM_{2.5}) and Particulate Matter 10 (PM₁₀); (see Abbreviations for definitions);
- deposited dust (the airborne dust mass settled on a land or water surface); and
- individual and complex mixtures of toxic air pollutants and similarly for odorous air pollutants.

A summary of criteria is provided in **Table 5-1**; these have been selected for their higher potential risk of air quality impact due to existing and potential future activities in the Precinct; for exhaustive lists, see NSW EPA (2022), and NSW EPA (2026).

Table 5-1 NSW EPA Air Quality Impact Assessment Criteria

Air pollutant	Averaging Period	Criterion	Relevance for Precinct
SO ₂	1h	75 ppb	Proximity to large coal-fired power stations: Eraring Power Station; Vales Point Power Station.
	24h	20 ppb	
NO ₂	1h	80 ppb	Higher risk pollutant for emissions from combustion processes including engine emissions; e.g. gas turbines, gas engines, diesel engines, road traffic.
	annual	15 ppb	
O ₃	8h	65 ppb	Higher risk of air quality impact due to existing high background levels, and increased by local and regional NO _x emissions.
PM _{2.5}	24h	50 µg/m ³	Higher risk of air quality impact due to relatively high background levels.
	annual	25 µg/m ³	
PM ₁₀	24h	25 µg/m ³	Higher risk of air quality impact due to relatively high background levels.
	annual	8 µg/m ³	
Deposited dust (insoluble solids)	annual	2 g/m ² /month (max. increase)	To assist with dust management e.g. during construction activities.
		4 g/m ² /month (max. total)	

These assessment criteria must be applied as follows (EPA, 2022):

1. At the nearest existing or likely future off-site sensitive receptor.
2. Incremental impact due to pollutant source alone for each pollutant must be reported in units and averaging periods consistent with impact assessment criteria.
3. Background concentrations must be included ("cumulative impact assessment").
4. Total impact (incremental impact plus background) must be reported as the 100th percentile in concentration or deposition units consistent with the impact assessment criteria and compared with the relevant impact assessment criteria.



5.6 Air Quality Guidelines

Air quality guidelines for consideration by air quality and odour management and assessment for proposals for the Precinct include:

- Transport for NSW, *Roadside air quality screening tool guideline* (2023), for the assessment of the operational impacts of road projects on air quality. The roadside air quality screening tool reflects model developments in NSW and other jurisdictions, designed to be consistent with the requirements of the Clean Air Society of Australia and New Zealand.
- NSW EPA *Local government air quality toolkit: guidance notes and visual guides*, including, but not limited to (EPA, 2026):
 - o Guidance notes:
 - Composite structural products.
 - Construction sites.
 - Neighbourhood smoke.
 - Climate change impacts.
 - o Visual guides:
 - Dust from construction sites.
 - Dust from small quarries.
 - Odour from food outlets.

6.0 Air Quality Study Area

6.1 Overview

As air pollutants disperse widely in the atmosphere, the air quality study area is not confined to the Precinct. This section summarises air pollution sources associated with the Precinct and immediate surroundings, local meteorology (focussing on winds), and existing air quality.

6.2 Sensitive Receptors

In terms of AQIA, some residential areas, isolated residences, and at least one school (“sensitive receptors” or “sensitive receivers”), are located very close, and even adjacent to, the Precinct boundaries; these are illustrated in the structure plan provided in **Appendix A**; and see the consideration areas illustrated in **Appendix B**.

Key receptor areas for consideration by the AQIA are:

- The residential-suburban areas Killingworth and Barnsley, and parts of Teralba south and adjacent of the Environmentally Constrained Area 6B.
- The five residential-rural areas including Wakefield in the south-west.



6.3 Existing Air Pollution Sources

6.3.1 Residual and Legacy Mining-Related Emissions

Areas associated with unrehabilitated land, tailings, stockpiles, haul routes, or disturbed surfaces may continue to generate dust during dry and windy conditions. Dust generation within these areas may constrain sensitive land uses; e.g., residential, schools, unless appropriately rehabilitated or buffered. Therefore, sensitive land uses should be directed away from areas with ongoing dust generation risk or staged until rehabilitation is demonstrably complete.

The existing Tailings Storage Facility (TSF) is undergoing rehabilitation, but with some parts remaining exposed, and is at a higher elevation than surrounding land. While rehabilitation is in progress, there is a risk of dust emissions from exposed areas due to wind erosion of exposed surfaces. Success of the rehabilitation program to stabilise and cover the surface with vegetation is very important, for the following reasons:

- Higher terrain elevations are exposed to increased wind speeds at higher elevations in the atmosphere.
- Higher wind speeds have greater land surface-eroding power leading to more dust emissions.
- Mine tailings surfaces can be disrupted by heavy rainfall, exposing surfaces for wind erosion.

6.3.2 Road and Rail Traffic

Existing and future road and rail traffic associated with the Precinct has the potential to influence land-use planning outcomes through road and rail vehicle air pollutant emissions, and amenity impacts at key interfaces with sensitive uses.

Current traffic movements related to mining, rehabilitation and industrial activity already contribute to local background conditions, while future redevelopment could result in increased construction traffic and longer-term changes in traffic volumes and patterns. These factors may constrain the siting and staging of sensitive land uses—particularly residential or recreational areas—near major access routes, haul roads or regional connections unless appropriately managed.

At a strategic level, traffic-related impacts do not preclude development within the Precinct, but support the need for thoughtful structure planning, including the use of buffers, transitional land uses, access management and staging to minimise exposure and maintain compatibility between future development and surrounding transport infrastructure.

6.3.3 Odour Sources

Two potentially significant odour sources are identified for the Precinct:

- Hunter Water Edgeworth Wastewater Treatment Works (WTW), located approximately 1,300 metres north-northeast of the Precinct's closest Developable Area DA1 (see **Appendix A**). While 1,000 metres is a reasonable separation distance for a WTW, the potential odour impacts on proposals in the northern parts of DA1 would need to be considered on a case-by-case basis.
- Concrush Waste Recycling Facility (WRF), located approximately 100-150 metres east of the easternmost parts of the Precinct's Developable Area DA1 (see **Appendix A**). Activities include recycling of "coarse forest mulch" by a shredding and screening process. While 500 metres would be a reasonable separation distance for



a WRF, the potential odour impacts on proposals in parts of DA1 would need to be considered on a case-by-case basis.

6.3.4 Other Air Pollution Sources

There are several nearby industrial and other activities that are potentially significant sources of air pollution; e.g., power generation, and regional industry, that contribute to the cumulative air emissions at a regional scale in the locality of the Precinct.

Two relatively close industrial activities that have potential to effect air quality in the Precinct are:

- Concrush Waste Recycling Facility (WRF) – considered in this section as a potential dust source. Activities include recycling of construction-and-demolition waste by a crushing and screening process. While 500 metres would be a reasonable separation distance for a WRF, the potential dust impacts on proposals in parts of DA1 would need to be considered on a case-by-case basis.
- Downer Asphalt Plant located approximately 800 metres south of Precinct boundaries, which should be an adequate separation distance. While NSW provides no separation distance for asphalt plants, the Environment Protection Authority Victoria (EPAV) provides a recommended separation distance of 1000 metres for asphalt plants with production greater than 100 tonne/week (EPAV, 2025).
- Teralba Quarry located approximately 900 metres of the southern boundaries of the Precinct, which should be an adequate separation distance – EPAV (2025) recommended separation is 500 metres for quarries where the main activities are: quarrying, crushing, screening, stockpiling and conveying of rock.

Most other air pollution sources in the region are sufficiently distant that air quality effects in the Precinct would be slight, or rare. For example, the main SO₂ sources are Eraring Power Station located approximately 14 km southwest of the Precinct, and the Vales Point Power Station located approximately 23 km south-southwest of the Precinct. There are other industrial sources in the Newcastle and Lower Hunter regions, but their air quality effects on the Precinct would be slight and hence are not assessed in this assessment.

6.4 Existing Air Quality

Air monitoring data are limited in and near the Precinct, as such a review of data generated by the EPA from three relatively close monitoring stations was undertaken:

- EPA Wyong on the Central Coast, located well to the south of the Precinct (results provided as a check);
- EPA Morisset, located approximately 18 km south-southwest of the Precinct, and probably the most representative location for the Precinct except for SO₂; and
- EPA Newcastle, located approximately 15 km east-northeast of the Precinct.

Summaries of monitoring data from the three EPA monitoring stations, and the corresponding NSW Impact Assessment Criteria (IAC), are provided in the following tables. Concentration units are parts per billion (ppb) by volume for the gaseous air pollutants, and microgram per cubic metre (µg/m³) for the particulate pollutants.



Table 6-1 Maximum 1-hour average NO₂ concentration (ppb)

Year	EPA Wyong	EPA Morisset	EPA Newcastle
2021	28	26	35
2022	31	25	38
2023	27	29	34
2024	26	25	33
2025	34	30	40

NSW EPA (2022) impact assessment criterion: 80 ppb

Table 6-2 Annual average NO₂ concentration (ppb)

Year	EPA Wyong	EPA Morisset	EPA Newcastle
2021	3	3	5
2022	2	2	4
2023	2	2	5
2024	3	2	5
2025	3.6	2.7	6.4

NSW EPA (2022) impact assessment criterion: 15 ppb

Table 6-3 Rolling 8-average O₃ concentration (ppb)

Year	EPA Wyong	EPA Morisset	EPA Newcastle
2021	50	58	48
2022	62	62	47
2023	56	58	58
2024	59	62	67
2025	42	85	51

NSW EPA (2022) impact assessment criterion: 65 ppb

Table 6-4 Maximum 1-hour average SO₂ concentration (ppb)

Year	EPA Wyong	EPA Morisset	EPA Newcastle
2021	30	71	37
2022	45	183	49
2023	44	86	44
2024	34	86	45
2025	42	85	51

NSW EPA (2022) impact assessment criterion: 100 ppb



Table 6-5 Annual average SO₂ concentration (ppb)

Year	EPA Wyong	EPA Morisset	EPA Newcastle
2021	5	16	6
2022	7	18	5
2023	5	11	6
2024	7	15	8
2025	7	10	6

NSW impact assessment criterion: 20 ppb

Table 6-6 Maximum 24-hour PM₁₀ concentration (µg/m³)

Year	EPA Wyong	EPA Morisset	EPA Newcastle
2021	44.9	28.5	44.3
2022	27.4	21.2	43.7
2023	37.8	33.8	49.0
2024	43.7	34.1	49.8
2025	52.7	64.2	66.0

NSW impact assessment criterion: 50 µg/m³

Table 6-7 Annual PM₁₀ concentration (µg/m³)

Year	EPA Wyong	EPA Morisset	EPA Newcastle
2021	13.5	9.6	19.2
2022	11.7	7.9	17.5
2023	13.9	11.2	20.1
2024	15.0	11.4	21.0
2025	14.8	11.6	21.7

NSW impact assessment criterion: 25 µg/m³

Table 6-8 Maximum 24-hour PM_{2.5} concentration (µg/m³)

Year	EPA Wyong	EPA Morisset	EPA Newcastle
2021	14.8	11.7	21.1
2022	11.5	11.2	18.7
2023	15.9	23.2	21.6
2024	15.6	17.9	22.5
2025	28.0	51.0	39.0

NSW impact assessment criterion: 25 µg/m³



Table 6-9 Annual PM_{2.5} concentration (µg/m³)

Year	EPA Wyong	EPA Morisset	EPA Newcastle
2021	4.7	3.9	6.3
2022	4.0	2.7	5.5
2023	4.9	5.0	6.7
2024	4.9	5.0	6.4
2025	5.1	5.0	6.5

NSW impact assessment criterion: 8 µg/m³

The following analysis identifies key points relating to the existing air quality surrounding the Precinct:

- NO₂: Newcastle concentrations would be higher due to the city's road traffic fleet at a level not seen across the Precinct. Based on the similarity of land uses compared to those surrounding the Precinct the Wyong and Morisset results indicate NO₂ concentrations in the Precinct will be less than half the IACs for NO₂.
- O₃: concentrations are 'moderately high' with a risk of air quality impact during events such as bushfires. Regional emissions of NO_x contribute to the relatively high O₃. The results emphasise the importance of minimising NO_x emissions across the Lake Macquarie and Newcastle regions, including the Precinct, and across the entire Lower Hunter.
- SO₂: concentrations are relatively high for Morisset due to emissions from the Eraring Power Station and to a lesser extent the more distant Vales Point Power Station. The Morisset results would not be representative of the Precinct. SO₂ concentrations for the Precinct could be expected to be slightly less than half the IACs.
- Particulate Matter as PM₁₀ and PM_{2.5} are moderately high in relation to the IACs. In the Precinct, high concentrations exceeding IACs could be expected due to smoke from bushfires and controlled burns, and dry conditions raising dust (Glencore, 2025).

In summary, the key air pollutants considered to be most important for the assessment of specific future proposals, are expected to be PM₁₀, PM_{2.5}, NO₂ and SO₂. However, specific proposals for the Precinct would need to be investigated on a case-by-case basis.

6.5 Local Meteorology

This section provides a summary of local meteorology pertinent to the dispersion of air pollutants near ground-level, with a focus on winds. Usually, the accurate determination of winds is the most difficult task for AQIA.

The Precinct is located in a complex, coastal-riparian zone, with rising terrain to the west. Determination of wind fields and turbulence for the purposes of AQIA is complicated by these factors, and the variable land use.

The broader region's synoptic scale winds dominate the Precinct's strongest winds, which are north-westerly, primarily occurring in winter (Glencore, 2025; and see **Figure 6-1**). The wind roses shown in **Figure 6-1** were created from Bureau of Meteorology (BoM) hourly observations at Nobbys Signal Station over 2013-2025, which is located in a relatively exposed situation on the coast near Newcastle. These wind patterns are indicative of regional winds, but exclude some of the details of the region around northern Lake Macquarie, where the Precinct is located – mainly these aspects:



- The Precinct is sheltered by westerly winds to some extent by high terrain in the Sugarloaf State Conservation Area to the west of the site; e.g., the terrain rises steeply by approximately 200 metres in only half a kilometre (km), approximately 3.0 km west of Wakefield.
- Local sea breezes and lake breezes will dominate when synoptic scale (regional) winds are weaker, modified to some extent by Lake Macquarie's complex shorelines.
- Highly variable coastal and riparian shorelines further complicate the local winds.

"Surface roughness" is of relevance to local winds. The relatively large areas of forest within and surrounding the Precinct increase surface roughness significantly. Surface roughness increases air turbulence and the mixing of air near ground-level, therefore assists with the dispersion of air pollutants. Also, to some extent, trees can trap airborne particulate matter, especially the larger ("nuisance dust") particles. For these reasons, the existence of well-established forested areas in the Project area and surrounds is an advantage for air quality management.

There are no current, high quality, meteorological observations being undertaken in the Precinct and immediate surrounds. In preparation for industrial proposals, the Precinct would benefit from the installation of a high quality, "reference" meteorological station centrally located in the Precinct, clear of obstructions and their sheltering effects – potentially on the rehabilitated tailings area (Environmentally Constrained Area 6B; see **Appendix A**).



Figure 6-1 Wind Roses 2013-2025: BoM Newcastle Nobbys Signal Station



*The wind roses are indicative of regional (Newcastle) winds only – winds on and near the Precinct will be affected by the complex Lake Macquarie-coastal situation, including sea and lake breezes. High terrain to the west will shelter the Precinct from some of the westerly winds. Winter westerly winds are expected to remain dominant in the Precinct.



7.0 Opportunities and Constraints Analysis

7.1 Overview

This section sets out the qualitative AQIA of the opportunities and constraints presented by the Project, drawing on the Structure Plan, and what has been learnt about the Precinct.

7.2 Precinct Activities

7.2.1 Existing Activities

Existing activities in and surrounding the Precinct, in relation to AQIA, have been discussed in **Section 6.3**.

7.2.2 Potential Future Construction

For potential future proposals, construction dust has the potential to act as a short- to medium-term constraint on development within the Precinct, particularly where large-scale earthworks, bulk material handling, rehabilitation works or haulage activities occur in proximity to existing or future sensitive receptors.

Construction dust impacts are most likely to arise during dry and windy conditions, having the potential to impact human health, and also on amenity through increased particulate deposition, if not properly managed. While these impacts are generally temporary and controllable, they can influence the timing, sequencing and layout of development, particularly for sensitive receptors such as residential areas and isolated residences, schools, and recreational spaces. As a result, construction activities may need to be staged appropriately, supported by buffer zones (separation), and/or subject to specific mitigation measures; e.g., progressive rehabilitation, surface stabilisation, vehicle management, and weather-based controls.

Construction dust does not preclude development within the Precinct, but represents a consideration that may constrain where and when certain land uses are delivered unless appropriate management and separation measures are embedded into future planning controls and development frameworks.

7.2.3 Potential Future Operations

The potential future air pollution sources in (and near) the Precinct are unknown. Relevant examples of potential future operations from the Master Plan, are listed in the following points – in all cases the pollutants for assessment will depend on specific fuel types and air emissions controls. These are relevant for the Developable Areas DA1 (Cockle Creek), DA2 (Killingworth), and DA3 (West Lake); with reference to the structure plan and consideration areas provided in **Appendix A** and **Appendix B**:

- General industrial – DA1 and DA2:
 - o Manufacturing, processing, fabrication, construction materials, heavy industry.
- Logistics and warehousing (including new road and rail alignments) – DA1 and DA2:
 - o Distribution centres, warehousing, freight and logistics operations.
- Light industrial – DA1, DA2 and DA3:
 - o Small-scale manufacturing, trade services, vehicle services, building supplies.
- Resource recovery and circular economy – DA1 and DA2



- o Waste processing, recycling, materials recovery, industrial symbiosis.
- Renewable energy generation and storage – DA1 and DA2
 - o Solar farms, battery energy storage systems, microgrids
- Advanced industry and technology – DA1, DA2 and DA3:
 - o Data centres, advanced manufacturing, research and development.

Major activities expected to be excluded from the Precinct due to the combination of higher risk of air quality impact, proximity of sensitive receptors, and existing (background) air pollutant concentrations, include:

- Open-cut mining due to emissions of PM₁₀ and PM_{2.5}.
- Larger, fossil-fuel power generation primarily due to emissions of NO_x and PM_{2.5}.
- Waste-to-energy plant due to many emitted substances and their uncertainties.
- Fuel farms due to the emissions of hydrocarbons while venting.

However, from the perspective of AQIA, any single proposal could be tested by air pollutant dispersion modelling in accordance with the approved methods (EPA, 2022; EPA, 2026).

7.3 Primary Constraints

7.3.1 Overview

In terms of AQIA, the primary constraints identified for the Precinct are:

- (1) The proximity of some sensitive receptors, especially those adjacent to Precinct boundaries (**Section 2.2**);
- (2) NSW air pollution impact assessment criteria (**Section 5.5**);
- (3) Existing air quality in the Precinct estimated from EPA-measured air pollutant concentrations (**Section 6.4**).

Air pollutant emissions at the source are not expected to be a constraint provided they meet the requirements of the *POEO (Clean Air) Regulation 2022* (**Section 5.3**). To achieve this, basic tenets for proponents are to use modern equipment employing “best global practice” emissions controls, (as far as is economically practicable and reasonable), and to keep equipment well maintained.

7.3.2 Protection of Sensitive Receptors

Sensitive land uses should be located and designed to avoid air quality and odour impacts in accordance with the NSW Environmental Protection Authority (EPA) *Approved Methods for the Modelling and Assessment of Air Pollutants in New South Wales* (EPA, 2022; EPA, 2026).

Where avoidance of air pollutant emissions is not possible, and where those emissions are judged to have the potential for significant air quality and/or odour impacts, detailed assessment and mitigation is required.

- “Detailed assessment” means meteorological and air quality and/or odour dispersion modelling in accordance with EPA (2022).
- The judgement of the potential for “significant” air quality and/or odour impacts is expected to be conducted by an experienced and appropriately qualified air quality



professional; e.g., a Clean Air Society of Australia and New Zealand (CASANZ) Certified Air Quality Professional (CAQP).

- As air quality/odour impact assessment is almost always complex, a proposal for the Precinct would be considered on a case-by-case basis to determine whether meteorological and air/odour dispersion modelling is required, and whether air monitoring is required. Factors for consideration would include proposal type, scale, and proximity to receptors.

Industrial and freight-intensive uses must manage dust and other emissions consistent with regulatory requirements at the Development Application stage; e.g., it is expected dust or construction air quality management plans and operational environmental management plans would be developed for larger proposals.

7.4 Opportunities

7.4.1 Overview

The cessation of coal mining operations presents a clear opportunity to improve air quality in and surrounding the Precinct, enabling more sensitive and diverse land uses.

With appropriate rehabilitation, planning, and design, the Precinct can continue to transition from a net air pollutant source to improved outcomes for the environment, leading to:

- Lower chronic exposure to PM_{2.5} and PM₁₀; and
- Improved liveability outcomes (amenity).

This section sets out several opportunities from the perspective of AQIA.

7.4.2 Rehabilitation of Mining Areas

The key opportunity for the Precinct is improved air quality relative to the (expected) current baseline PM₁₀, PM_{2.5} and deposited dust levels due to rehabilitation of the tailings. As mentioned earlier in this assessment (**Section 6.3.1**), success of the rehabilitation program to stabilise and cover the tailings with rocky material, crusted surfaces, and/or vegetation, is very important.

- **Recommendation:** an airborne particulate / deposited dust monitoring program in the Precinct may be able to demonstrate improved mitigation of tailings dust emissions.

More generally, revegetation and stabilisation of exposed surfaces in the Precinct will reduce the potential for dust emissions. Vegetation acts as a wind break, dust sink, and assists with erosion control. Use of native vegetation should increase resilience.

7.4.3 Environmentally Constrained Areas as Buffer Zones

The Environmentally Constrained areas (see **Appendix A**), providing vegetated corridors between the new land uses and air emission sources such as industries and roads, can reduce dust transport and improve air pollutant dispersion (see also **Section 6.5**). Benefits will exist for amenity, biodiversity and air quality.

7.4.4 Low Air Emission Industries

The integration of low air emission industries such as renewable energy generation, low-emission industrial uses, and operations with electrified vehicle fleets and equipment, would



represent a shift from high emissions to low or very low air emissions land use, creating a “clean air Precinct”.

7.4.5 Air Quality Monitoring Legacy

There is an opportunity to lever off long-term air/dust monitoring datasets to validate improved post-closure conditions, supporting planning approvals.

- **Recommendation:** If possible, legacy and/or existing air monitoring infrastructure could be used to establish a new monitoring network, providing a new baseline for future development in the Precinct.

7.4.6 Co-Benefits: Air Quality and Greenhouse Gases

Post-mining land uses selected to deliver lower air pollutant emissions such as NO_x and particulate matter, will lower greenhouse gas emissions also. Examples are:

- Solar farm vs. fossil fuel-power generation; and
- Maintenance and re-forestation improves air pollutant dispersion, is a dust sink, and (to a small extent given the relatively small areas), is a carbon dioxide (greenhouse gas) sink.

7.4.7 Designing for Future Air Quality Resilience

The Structure Plan and future proposals for the Precinct should give consideration to the following future climate impacts from global warming:

- Potential for more days of bushfire smoke, with global warming expected to lead to longer bushfire seasons.
- More frequent and more intense wind, dust, and precipitation storms.

Future-proofing would include consideration given to building orientation, “shelter belts”, and particulate filtration in sensitive buildings such as child care centres, schools, and medical centres.

7.5 Assessment of Structure Plan

7.5.1 Assessment of Sub-Precincts

Assessments for each of the sub-precincts illustrated in the Precinct Structure Plan, (see **Appendix A**), and summaries of each sub-Precinct’s characteristics and typology (land use), are provided in **Table 7-1** (from DPHI, 2026).

7.5.2 Assessment of Teralba Southgate

Teralba Southgate encompasses land adjacent to Teralba local centre, characterised by proximity to public transport, the heritage-listed Teralba Environmentally constrained Area, and established residential and commercial parts of Teralba village.

Teralba Southgate is envisaged to accommodate medium-density residential, activated commercial and other community uses.

There is a low risk of air quality impact from these activities – construction projects for this relatively small area, in close proximity to sensitive receptors, will require careful dust and odour impact management.



Table 7-1 Structure Plan Assessment

Sub-Precinct	Characteristics	Land Use (Typology)	Air Quality and Odour Assessment
DA1 Cockle Creek	Characterised by legacy industrial infrastructure and the presence of the TSF, also encompassing mine rail loop and associated legacy operational infrastructure. Parts subjected to extensive industrial activity.	General industrial, manufacturing, logistics, warehousing and freight-related uses. Low-frequency rail freight including rail loop.	Sensitive receptors on northern and SE boundaries; SE protected by environmentally constrained area. Construction projects will require dust and odour impact management. Industrial proposals would need to be assessed on a case-by-case basis. AQIA results will depend on risks associated with air emissions substances and amounts emitted; mitigation by use of "best practice" equipment, maintenance, air emissions controls, and stack heights where applicable. Proposals involving increased frequency of rail freight movements on rail loop and future Lower Hunter Freight Corridor would require case-by-case assessment. The intermittent nature of locomotive emissions would not be expected to cause a significant air quality impact. However, there may be specific receptor cases such as pedestrian bridges over rail lines that would require some further investigation. A risk is associated with the TSF covering potentially breaking up or if unable to maintain rehabilitated vegetation, in which case could become a dust source due to wind erosion of exposed area. Note the revegetation will need to avoid disrupting the TSF's protective layers. Data centre may be satisfactory assuming high stacks on high infrastructure (buildings), noting air quality risks of data centres from engine operations due maintenance and emergency use.
DA2 Killingworth	Near established residential areas of Killingworth and Barnsley. Relatively accessible topography with existing road connections via Wakefield Road and The Broadway.	Light industrial, energy intensive uses, technology uses, small-medium employment.	Nearby sensitive receptors to west and northeast boundaries. Construction projects will require dust and odour impact management. Industrial proposals would need to be assessed on a case-by-case basis. AQIA results will depend on risks associated with air emissions substances and amounts emitted; mitigation by



Sub-Precinct	Characteristics	Land Use (Typology)	Air Quality and Odour Assessment
	Interface with environmentally constrained areas and biodiversity corridors to south and east. Previously disturbed land associated with former mining operations may reduce environmental constraints and enable short term development.		use of "best practice" equipment, maintenance, air emissions controls, and stack heights where applicable. Data centre may be satisfactory assuming high stacks on high infrastructure (buildings), noting AQ risks of DCs due to diesel/gas engine operations due maintenance and emergency use.
DA3 West Lake	Encompasses former Westside Open Cut void, which is now a water-filled lake, and surrounding rehabilitation areas, characterised by environmental and landscape values. Conservation and recreation asset that complements and enhances the employment-focussed sub-precincts.	Conservation, recreation, nature-based tourism and complementary low-impact uses. Likely a later-stage activation area, contingent upon confirmation of access feasibility, servicing viability and remediation requirements.	Very low risk of air quality impact. Any construction projects will require dust and odour impact management. Any proposed activities associated with fire trails and associated infrastructure, especially for unpaved roads, should consider air quality risks; e.g., due to dust emissions from unpaved roads and other exposed areas.
DA5 Barnsley West	Characterised by relative isolation, constrained access, existing transmission lines and interface with rural landscape and conservation areas.	Light industrial/commercial	Low air quality risk. Construction projects will require dust and odour impact management. Industrial proposals would need to be assessed on a case-by-case basis. AQIA results will depend on risks associated with air emissions substances and amounts emitted; mitigation by use of "best practice" equipment, maintenance, air emissions controls, and stack heights where applicable.
6A Wakefield 6B Central 6C Barnsley, west 6E Teralba	Encompass tracts of remnant native vegetation, rehabilitation areas, flood prone land, waterways and ecological corridors.	Environmentally constrained Areas protected and managed as integral elements of Precinct's environmental framework.	Very low risk of air quality impacts. Any proposed activities associated with fire trails and associated infrastructure, especially for unpaved roads, should consider air quality risks; e.g., due to dust emissions from unpaved roads and other exposed areas.



Sub-Precinct	Characteristics	Land Use (Typology)	Air Quality and Odour Assessment
	Conservation network supports biodiversity, water cycle management, and landscape identity including Connecting with Country.	Distinctive landscape settings enhancing amenity.	Note there may be existing odour impacts from wetlands, which would need to be addressed by any odour impact assessment.
6D Northern Peninsula			Very low risk of air quality impacts. Any proposed activities associated with fire trails and associated infrastructure, especially for unpaved roads, should consider air quality risks; e.g., due to dust emissions from unpaved roads and other exposed areas. Potential existing odour impacts from Edgeworth WTW, Concrush Waste Recycling Facility, and to some extent from riparian zones, would need to be considered for any (cumulative) odour impact assessment.
6F Wakefield Road			Very low risk of air quality impacts. Any proposed activities associated with fire trails and associated infrastructure, especially for unpaved roads, should consider air quality risks; e.g., due to dust emissions from unpaved roads and other exposed areas.



8.0 Outcomes

Precinct Overview – Air Quality and Odour

Local meteorology, terrain, and land use are important first considerations for air quality and odour impact assessment. The Precinct is complex, located between a coastal-riparian zone to the east, and rising terrain to the west. Determination of meteorological parameters is complicated by the factors:

- The broader region's synoptic scale winds are expected to dominate annual wind patterns. However, some areas will be affected by high terrain associated with the Sugarloaf State Conservation Area to the west.
- Local sea breezes and lake breezes will dominate when synoptic scale winds are weaker, modified by the complex shorelines.

Key air pollutant indicators for specific proposals in the Precinct will be determined on a case-by-case basis, but are expected to include at least:

- Airborne particulate matter (PM₁₀ and PM_{2.5}).
- Gaseous NO₂.
- Odour – complex mixtures

Factors affecting air pollutant dispersion are:

- Some sensitive receptors are adjacent to Precinct boundaries – careful consideration will be needed for separation distances between proposed industries and sensitive receptors; again, these should be tested on a case-by-case basis:
 - o Larger-scale industries such as data centres, resource recovery and heavy industry are likely to require detailed (modelling) air quality impact assessment undertaken in accordance with the Approved Methods; depending on the air pollutant sources identified.
 - o Smaller-scale industries such as an opto-electronics manufacturing centre, may require a qualitative air quality impact assessment only; depending on the air pollutant sources identified
- The Precinct's decommissioned tailings storage facility, currently undergoing rehabilitation, is at a higher elevation than surrounding land. While rehabilitation is in progress there is a risk of dust impacts due to wind erosion of exposed surfaces due to:
 - o Higher terrain elevations being exposed to the higher wind speeds that exist higher up in the atmospheric mixing layer.
 - o Higher wind speeds have greater land-eroding power leading to greater dust emissions.
 - o Tailings surfaces can be disrupted by heavy rainfall, re-exposing the surfaces available for wind erosion.
- Therefore, success of the rehabilitation program to stabilise and cover the tailings with rocky material, crusted surfaces, and/or vegetation, is very important.
- Relatively large areas of forest within and surrounding the Project area enhance the mixing of air, assisting with air pollutant dispersion – an advantage for air quality management.



Expectations for buffers are:

- Separation distances between potential air pollution sources and sensitive receptors will need to be such that air quality impact assessment criteria are met at the sensitive receptor locations specified in NSW EPA (2022); noting the potential updates detailed in NSW EPA (2026).

Aims

The aims for air quality and odour management are to:

- Maintain air quality and amenity for people within the Project area and surroundings.
- Minimise air pollution impacts on flora and fauna within the Project area and surroundings.
- Ensure that development within the Project area minimises air quality and odour impacts.
- Ensure air quality, odour, and air pollutant deposition monitoring for developments are appropriate for construction, operations, and decommissioning activities.

Performance criteria

Air quality and odour impact assessment criteria are listed in **Table 8-1**.

Table 8-1 Precinct Performance Criteria – Air Quality and Odour

Performance Criterion	Intent
Air emissions limits and other requirements prescribed by the NSW POEO (Clean Air) Regulation 2022.	Air pollutant emissions from point (stack) sources should not exceed concentration limits prescribed by the NSW POEO (Clean Air) Regulation 2022. The limits are based on “levels achievable through the application of”, as stated by the NSW EPA, “reasonably available technology and good environmental practices”.
NSW EPA impact assessment criteria, by monitoring and/or modelling (NSW EPA, 2022; NSW EPA, 2026).	Manage significant air pollutant emissions to maintain human health by avoiding and minimising higher air pollutant concentrations, as determined by, (where needed), comparisons with NSW EPA impact assessment criteria, by monitoring and/or modelling (NSW EPA, 2022; NSW EPA, 2026).
	Odour emissions management to maintain amenity by avoiding and minimising higher odour concentrations as determined by (where needed), comparisons with NSW EPA odour assessment criteria, by monitoring and/or modelling (NSW EPA, 2022; NSW EPA, 2026). See also amenity performance criterion.
Very few amenity complaints due to dust, odour, visible plumes.	Control of dust and odour emissions to maintain amenity by: (1) strong engagement with affected communities, and, where needed: (2) monitoring; and (3) assessment.
World Health Organization (WHO) standards for air pollutant deposition.	Minimise the deposition of gaseous and particulate air pollutants to maintain flora and fauna health on land, riparian-coastal areas, and on the sea.



Recommendations

All future Development Approvals, re-zoning, and change of land use are to be accompanied by proportional and site-specific air quality and/or odour impact assessments as required.

Additionally, the success of the rehabilitation program to stabilise and cover the tailings with rocky material, crusted surfaces, and/or vegetation, is very important:

- An airborne particulate / deposited dust monitoring program in the Precinct may be useful to demonstrate improved mitigation of tailings dust emissions.
- More generally, revegetation and stabilisation of exposed surfaces in the Precinct will reduce the potential for dust emissions – vegetation acts as a wind break, a dust sink, and assists with erosion control. Use of native vegetation should increase resilience of exposed surfaces to wind erosion.

There is an opportunity to lever off long-term air / dust monitoring datasets to validate improved post-closure conditions, supporting planning approvals:

- If possible, legacy and / or existing air monitoring infrastructure could be used to establish a new monitoring network, providing a new baseline for future development in the Precinct.

9.0 Conclusion

An Air Quality and Odour Assessment to inform planning considerations associated with the proposed rezoning of the Precinct was undertaken. This report sets out the assessment methods for determining air quality and odour performance criteria for construction and operations within the Precinct, allowing for the investigation and development of air emissions control strategies during the planning and design phases.

Future development of the Precinct will need to minimise environmental impacts due to air pollutant emissions. Air quality impacts within the Precinct can be effectively managed by, where needed, monitoring and detailed (modelling) investigations on a case-by-case basis, and environmental (air quality) management plans for larger projects.

In addition (**Section 7.0**):

- The rehabilitation program to stabilise and cover the tailings with rocky material, crusted surfaces, and/or vegetation, is very important. More generally, revegetation and stabilisation of exposed surfaces in the Precinct will reduce the potential for dust emissions. Vegetation acts as a wind break, dust sink, and assists with erosion control. Use of native vegetation should increase resilience.
- The Environmentally constrained Areas will provide vegetated corridors between new land uses and air emission sources such as industries and roads, reducing dust transport and improving air pollutant dispersion.
- Integration of low emissions industries will represent a shift from high emissions to lower air emissions land use.
- Planning and design to prepare for future climate impacts from global warming such as, potentially, longer bushfire seasons, and more frequent and more intense wind, dust, and precipitation storms.
- Post-mining land uses selected to deliver lower air pollutant emissions for air quality management purposes, will lower greenhouse gas emissions also (a co-benefit).



This desktop AQIA was limited to a review of confidential Project material and readily available, publicly available information. Exclusions from the scope of works included:

- Meteorological, and air quality (pollutant) and odour monitoring.
- Air pollutant deposition monitoring.
- Meteorological and air pollutant dispersion modelling.

The AQIA is intended for strategic planning purposes. It does not replace project-specific, air quality and/or odour impact assessment that may be required in a later stage of development and/or approval.

All future Development Approvals, re-zoning, and change of land use are to be accompanied by proportional and site-specific air quality and/or odour impact assessments as required; also:

- An airborne particulate / deposited dust monitoring program in the Precinct may be useful to demonstrate improved mitigation of tailings dust emissions.
- More generally, revegetation and stabilisation of exposed surfaces in the Precinct will reduce the potential for dust emissions – vegetation acts as a wind break, a dust sink, and assists with erosion control. Use of native vegetation should increase resilience of exposed surfaces to wind erosion.
- If possible, legacy and / or existing air monitoring infrastructure could be used to establish a new monitoring network, providing a new baseline for future development in the Precinct.



References

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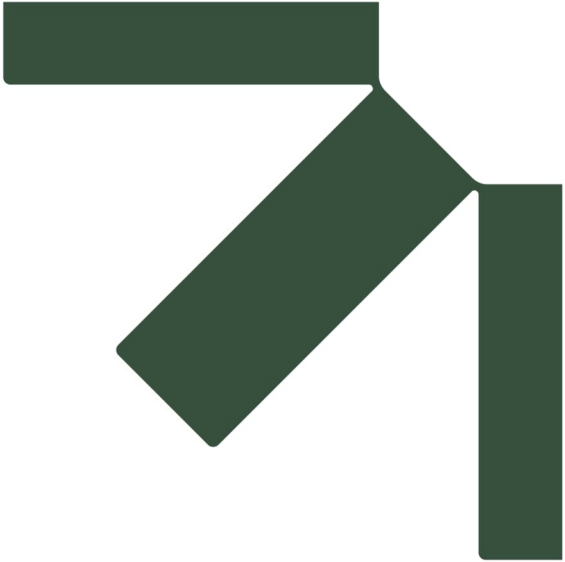
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Appendix A Structure Plan

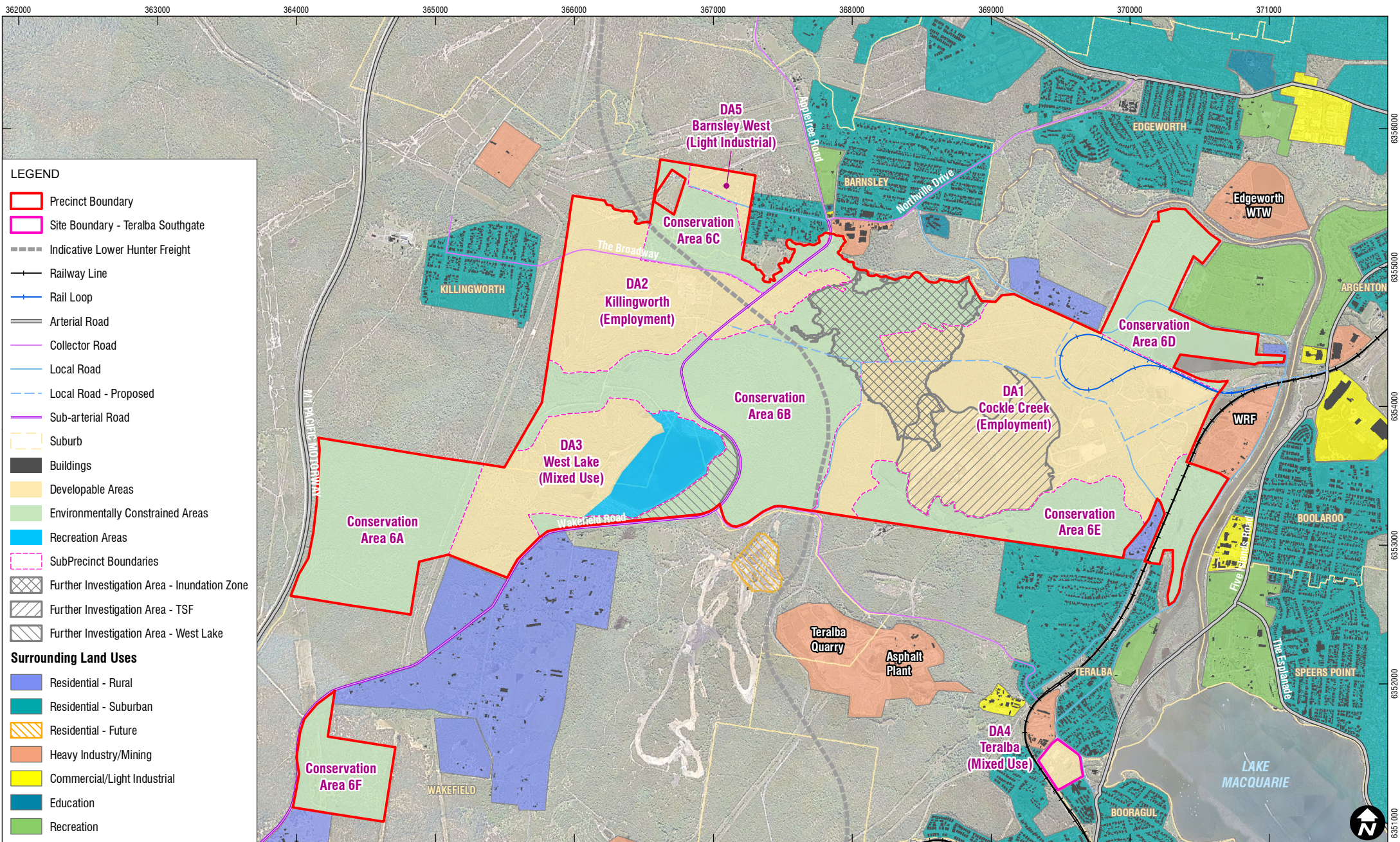
Air Quality & Odour Assessment

Post Mining Land Use Pilot Project – Macquarie Coal Complex (the Rezoning Proposal)

Department of Planning, Housing and Infrastructure

SLR Project No.: 630.032981.00001

1 July 2026



0 1 2 km

Scale: 1:35,000 at A4
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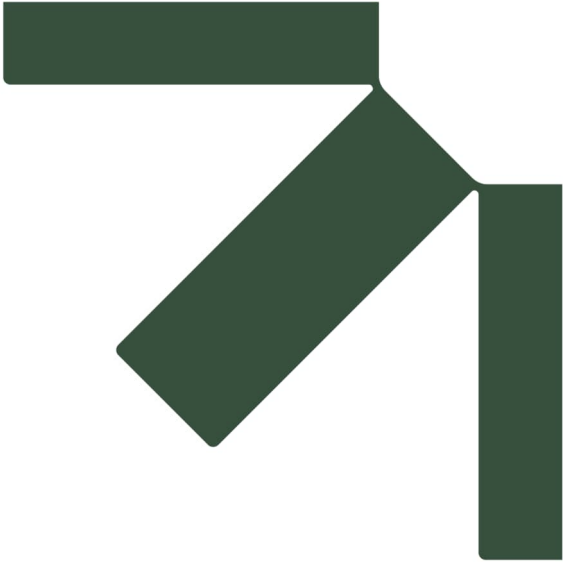
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Date Drawn: 01-Jul-2026
Project Number: 630.032981.00001



STRUCTURE PLAN Air Quality

FIGURE A1



Appendix B Consideration Areas

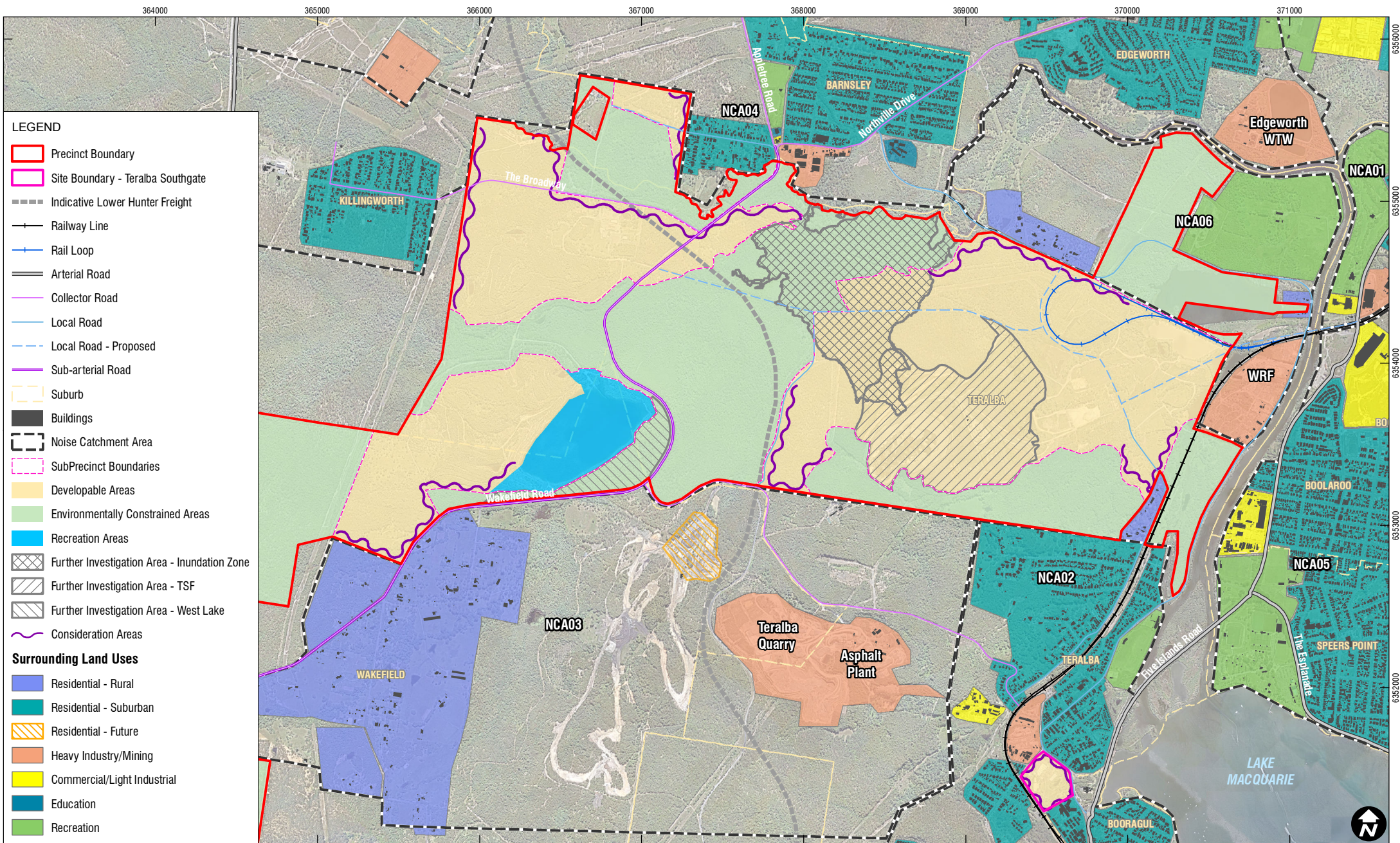
Air Quality & Odour Assessment

Post Mining Land Use Pilot Project – Macquarie Coal Complex (the Rezoning Proposal)

Department of Planning, Housing and Infrastructure

SLR Project No.: 630.032981.00001

1 July 2026



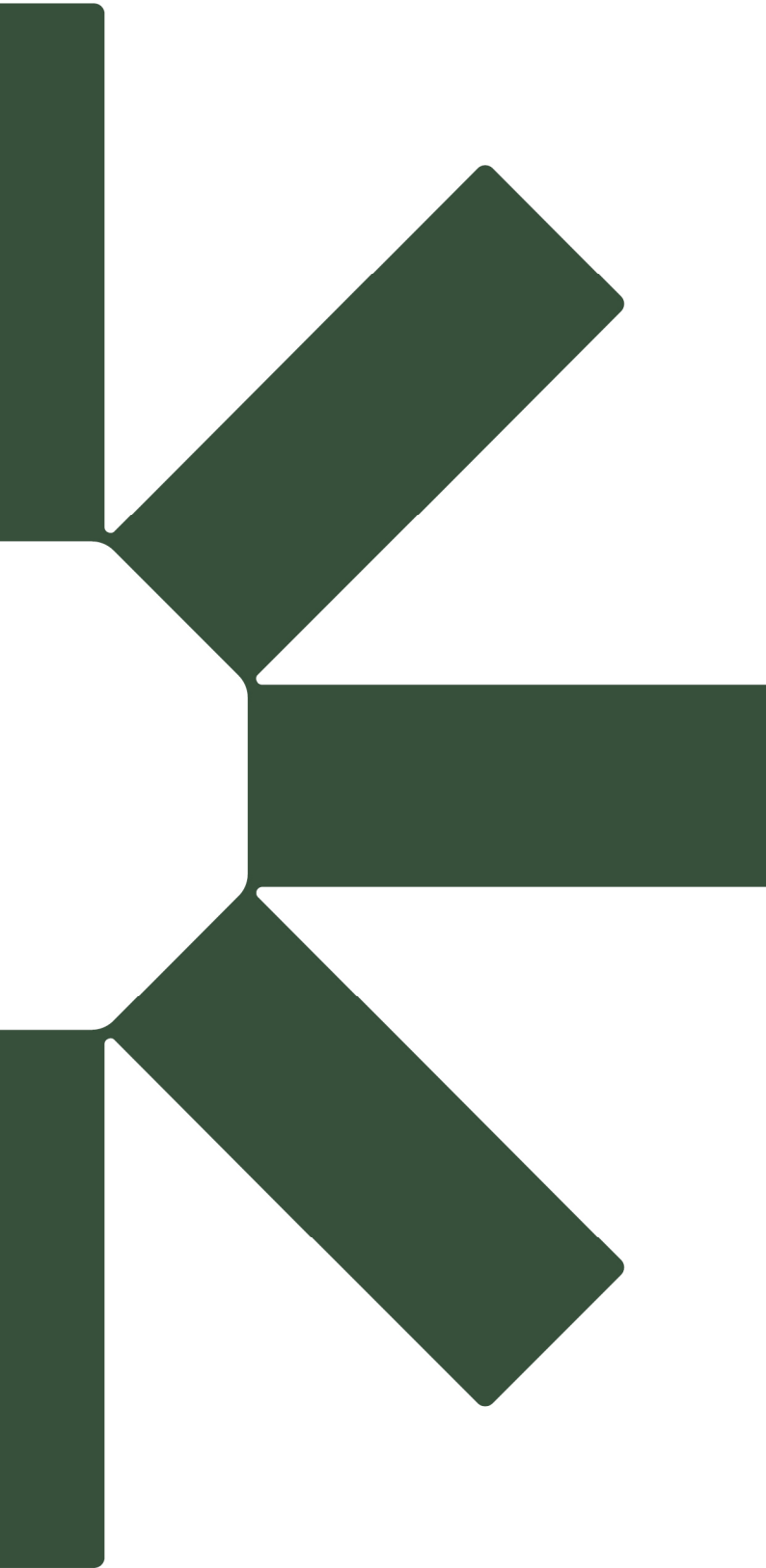
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Date Drawn: 01-Jul-2026
 Project Number: 630.032981.00001



CONSIDERATION AREAS

FIGURE B1



Making Sustainability Happen